

WORLD IN BRIEF

AI News Intelligence Platform

UML Modeling Assignment

01 Use Case Diagram

From Student Submission (image.png) & SRS Section 7.2, 8

02 Entity-Relationship (ER) Diagram

Extracted from SRS Section 4 (Data Requirements & Database Design)

03 Class Diagram

Derived from SRS Section 7.3 (UML Modeling Support)

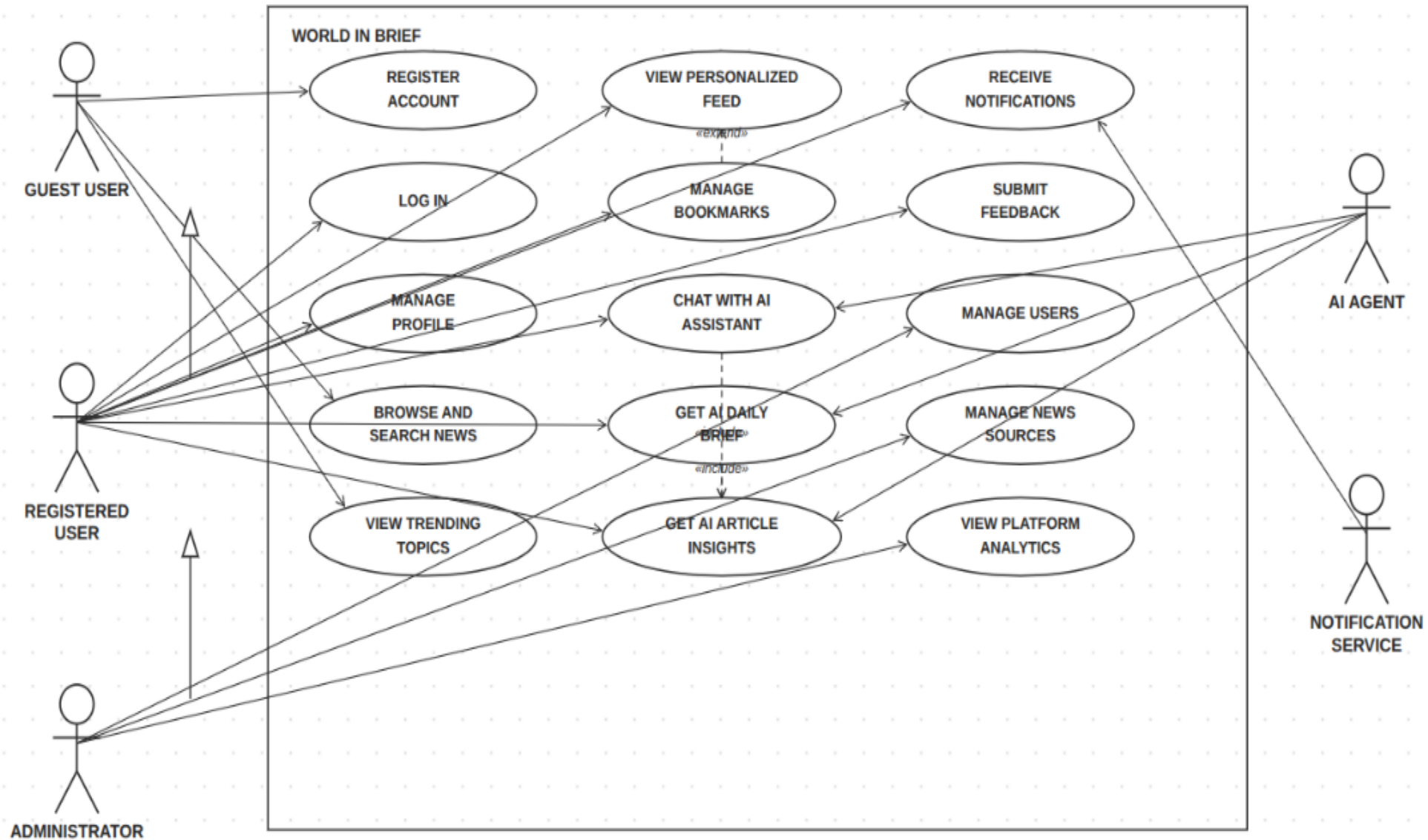
Software Requirements Specification (IEEE 830-1998)

Document Version 1.0 | July 16, 2026

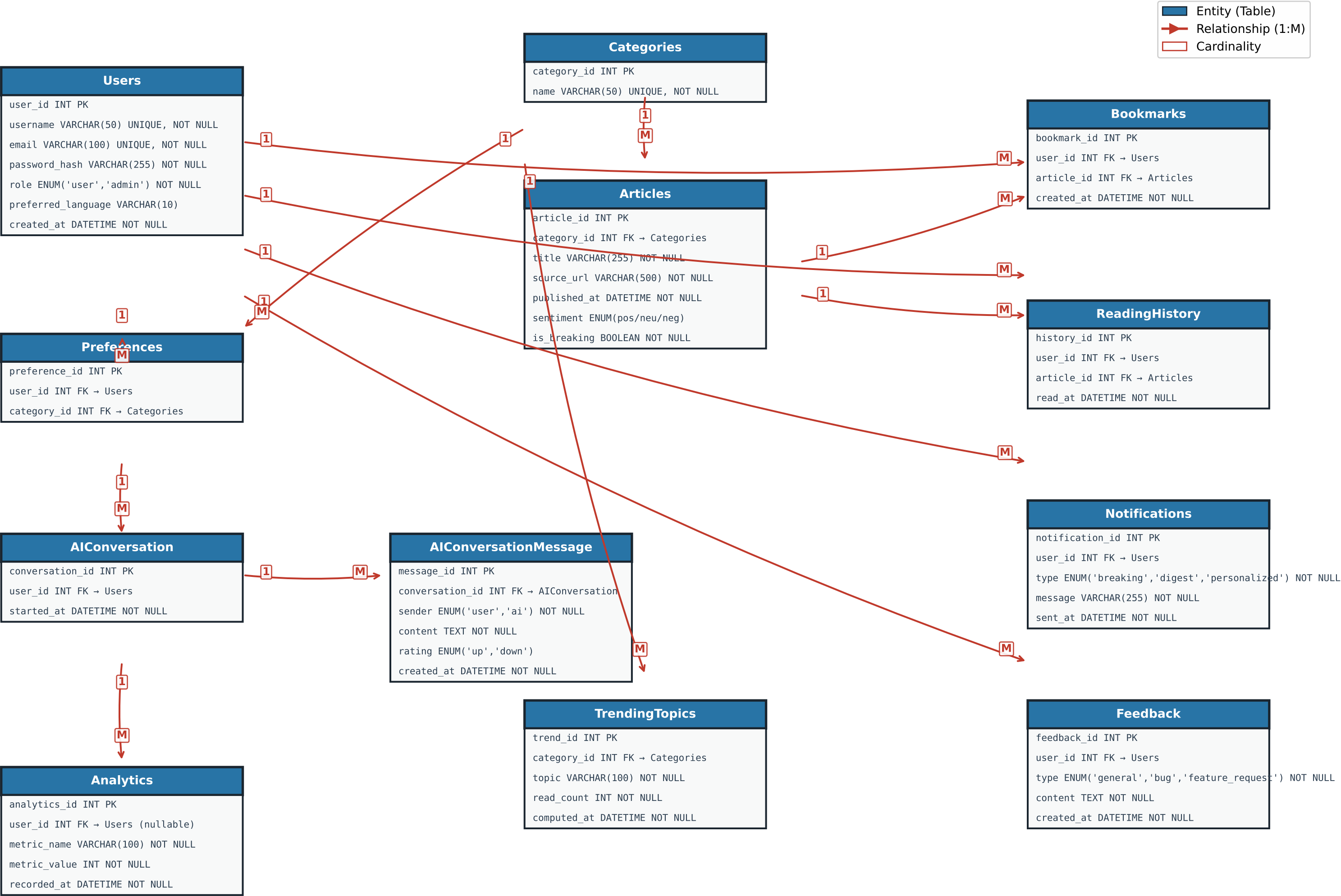
Prepared for Academic Submission — Software Engineering Course Project

USE CASE DIAGRAM | WORLD IN BRIEF — AI News Intelligence Platform

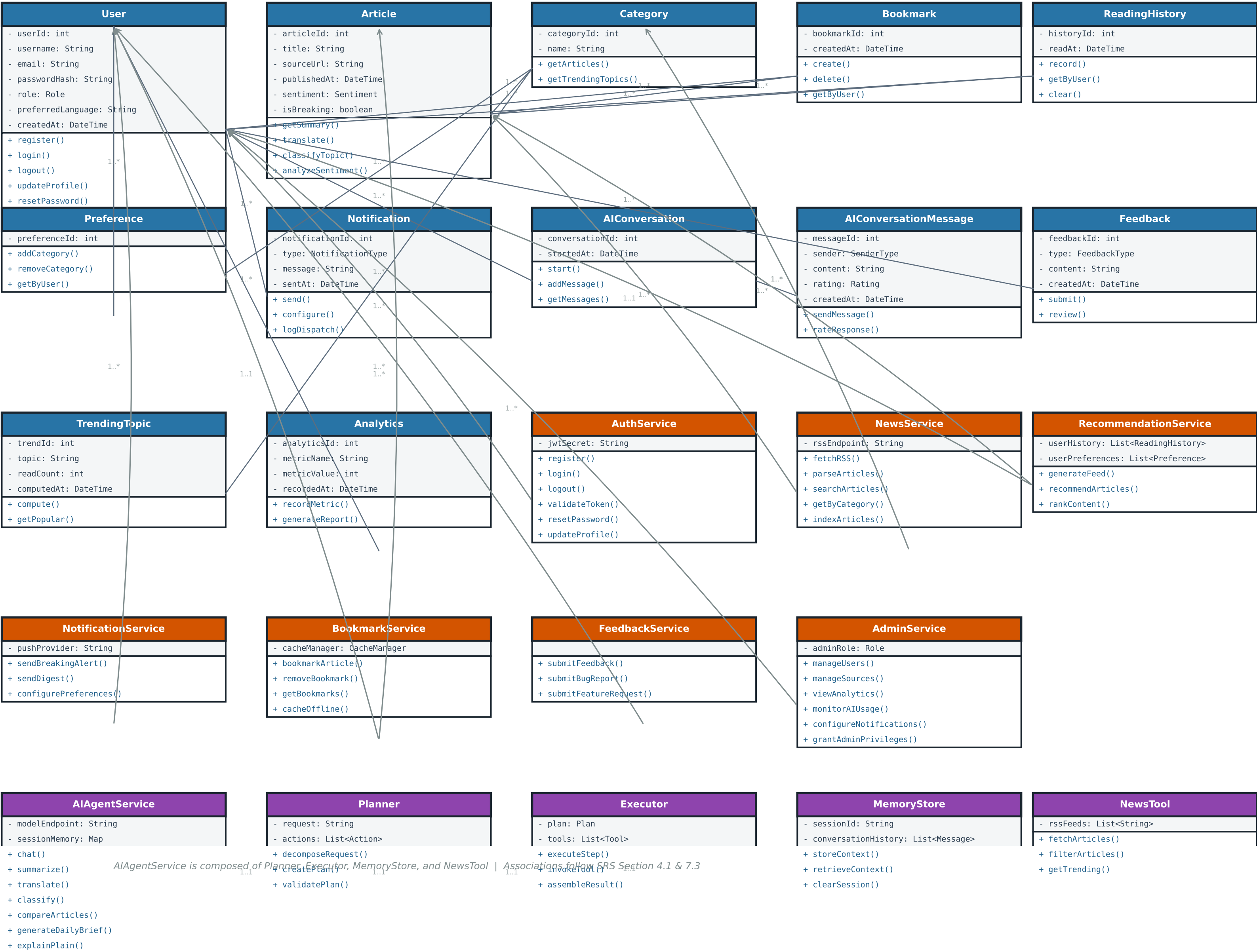
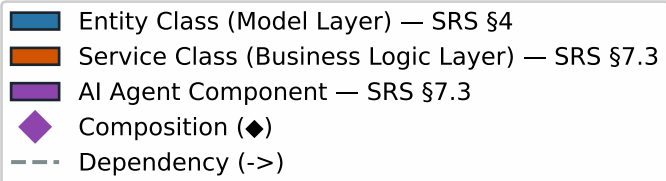
WORLD IN BRIEF - AI NEWS INTELLIGENCE PLATFORM



ENTITY-RELATIONSHIP (ER) DIAGRAM | WORLD IN BRIEF
Extracted from SRS Section 4 — Data Requirements and Database Design



CLASS DIAGRAM | WORLD IN BRIEF
Derived from SRS Section 7.3 — UML Modeling Support



SRS TRACEABILITY & REFERENCES

USE CASE DIAGRAM

- SRS Section 7.2 — Use Case Diagram (Actors & UC relationships)
- SRS Section 7.1 — Actors: Guest User, Registered User, Administrator, AI Agent, Notification Service
- SRS Section 8 — Use Cases UC-01 to UC-24 (Full catalog with preconditions, flows, postconditions)
- SRS Section 2.3 — User Classes and Characteristics

ENTITY-RELATIONSHIP DIAGRAM

- SRS Section 4 — Data Requirements and Database Design (12 entities with full field specs)
- SRS Section 4.1 — Entity Relationships (1:M mappings between all entities)
- SRS Section 4.2 — Normalization (3NF compliance, single-column surrogate PKs)
- Entities: Users, Articles, Categories, Bookmarks, ReadingHistory, Preferences,
- Notifications, AIConversation, AIConversationMessage, Feedback,
- TrendingTopics, Analytics

CLASS DIAGRAM

- SRS Section 7.3 — Class Diagram (Candidate classes & service-layer classes)
- SRS Section 7.3 — AIAgentService composition: Planner, Executor, MemoryStore
- SRS Section 7.8 — Component Diagram (Major components reference)
- SRS Section 11 — Software Architecture (Layered architecture: Presentation, API, Auth,
- Business Logic, AI Agent, Service, Repository, Database, Ollama)
- Model Classes: User, Article, Category, Bookmark, ReadingHistory, Preference,
- Notification, AIConversation, AIConversationMessage, Feedback,
- TrendingTopic, Analytics
- Service Classes: AuthService, NewsService, RecommendationService, AIAgentService,
- NotificationService, BookmarkService, FeedbackService, AdminService

This document combines the student-submitted Use Case Diagram with ER and Class Diagrams extracted directly from the World in Brief SRS (IEEE 830-1998, Version 1.0)

NON-FUNCTIONAL REQUIREMENTS TRACEABILITY

- NFR-20: Modular, layered architecture isolating changes to individual layers
- NFR-22: Each functional module exposes a clearly defined service boundary
- Schema normalized to 3NF (SRS Section 4.2) supporting NFR-20 maintainability